Focused Literature Review

Water Quality Prediction

Tree ensembles, feedforward neural networks and evaluation of a calculated index

Project 18 Team

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Scope

This is a focused review of the methods and target definition used in Project 18, rather than an exhaustive systematic review of water-quality research. The sources below connect directly to implemented choices. We separate theoretical motivation from observations produced by the submitted code.

The central issue is construct validity: learning a score calculated from the same predictors tests an emulator. A valid index definition and honest evaluation are necessary before comparing algorithm complexity.

Source access: October 2026.

1. Index definition and tree models

Karim et al. dataset [1]

The release combines eight harmonized chemistry parameters across five countries and provides both a numerical index and class bands. It enables a common regression target, but its supplied preprocessing has already filled missing values. The original observation flags and full conversion history are unavailable. We therefore preserve the original combined CSV in an archive and audit the released formula rather than inventing laboratory provenance.

CCME user manual [2]

The manual defines an index through scope, frequency and amplitude against chosen objectives over a defined assessment period. Objectives and temporal aggregation are part of the target definition. The released notebook's single-row, upper-only adaptation differs from that framework. This distinction limits environmental interpretation and motivates the exact-reconstruction check.

Random Forests [3]

Breiman's forest averages randomized trees to reduce prediction variance. It provides the assignment's classical tabular-model background. We select XGBoost as the implemented AML baseline; a random forest is discussed for context and is not presented as a trained competitor.

XGBoost [4]

Chen and Guestrin describe a scalable regularized boosting system. Sequential trees model residual structure and threshold interactions. In this project, bounded tree depths and early stopping control fitting. The source formula contains explicit thresholds, making this a plausible model choice before viewing the test results.

2. Neural networks and reliability

Deep feedforward networks [5]

A multilayer perceptron composes learned affine transformations with nonlinear activations. This motivates a small regression network for interacting sensor measurements. Scaling improves optimization, while a linear output predicts a continuous quantity. Tabular columns do not form a temporal sequence, so an LSTM is not justified by their order.

Tabular model comparisons [6]

Grinsztajn and colleagues benchmark tree ensembles and neural models on tabular problems and analyze their differing inductive biases. Their findings motivate testing a strong tree baseline before claiming an advantage for deep learning. Their benchmark does not predict which model must win on this dataset; our results remain separate empirical evidence.

Dropout [7]

Srivastava and colleagues propose dropping units during training to reduce co-adaptation. We use dropout 0.1 with weight decay and validation stopping. This project does not include a dropout ablation, so its individual contribution cannot be isolated from the other regularizers.

Conformal prediction [8]

Angelopoulos and Bates explain uncertainty sets calibrated using held-out errors under exchangeability. We use the corrected residual quantile as a transparent interval method. Because chronology introduces dependence and shift, we assess observed coverage and avoid promising distribution-free validity for this particular holdout.

Resulting research gap

The present dataset supports a reproducible methods comparison, but independent quality outcomes and original measurement flags remain missing. The next contribution should improve target independence and evaluation design before adding network depth.

3. Evidence-to-method mapping

Source	Implemented decision	Limit of inference
Dataset + CCME [1,2]	One released numerical target and exact formula audit	No claim of official site-period assessment
XGBoost [4]	Regularized depth-4/depth-6 search	Bounded search and one seed
Feedforward networks [5]	Small ReLU MLP, transformed inputs	No guarantee of DL superiority
Tabular benchmark [6]	Strong classical baseline before DL	External results do not replace local evaluation
Dropout [7]	Dropout plus AdamW and early stopping	No isolated regularizer ablation
Conformal overview [8]	Held-out residual intervals	Temporal and subgroup coverage can fail

References

[1] Karim et al. A Comprehensive Surface Water Quality Monitoring Dataset (1940-2023), Figshare v2, 2025. https://doi.org/10.6084/m9.figshare.27800394.v2. CC BY 4.0.

[2] Canadian Council of Ministers of the Environment. CCME Water Quality Index User's Manual, 2017 update. https://ccme.ca/en/res/wqimanualen.pdf.

[3] Breiman, L. Random Forests. Machine Learning 45, 5-32, 2001. https://doi.org/10.1023/A:1010933404324.

[4] Chen, T., and Guestrin, C. XGBoost: A Scalable Tree Boosting System. KDD, 2016. https://doi.org/10.1145/2939672.2939785.

[5] Goodfellow, I., Bengio, Y., and Courville, A. Deep Learning. MIT Press, 2016. Chapter 6. https://www.deeplearningbook.org/contents/mlp.html.

[6] Grinsztajn, L., Oyallon, E., and Varoquaux, G. Why do tree-based models still outperform deep learning on tabular data? 2022. https://arxiv.org/abs/2207.08815.

[7] Srivastava, N. et al. Dropout: A Simple Way to Prevent Neural Networks from Overfitting. JMLR 15(56), 1929-1958, 2014. https://jmlr.org/papers/v15/srivastava14a.html.

[8] Angelopoulos, A. N., and Bates, S. A Gentle Introduction to Conformal Prediction and Distribution-Free Uncertainty Quantification. 2021, revised 2022. https://arxiv.org/abs/2107.07511.